

Ex. No: 3 Rule Based Morphological Analyzer

Implement Rule Based Morphological Analyzer for the given Datasets

Datasets:

- 1. Brown Corpus
- 2. Reuters Corpus
- 3. Newsgroups Corpus
- 4. Universal Dependencies English Tree Bank

Aim

To implement a Rule-Based Morphological Analyzer that applies hand-crafted linguistic rules to identify roots, prefixes, suffixes, and morphological categories (inflectional and derivational) for words in the given corpora.

Procedure/Pseudocode

- 1. Load the corpus and extract the vocabulary (unique word list).
- 2. Define morphological rules as dictionaries mapping suffixes/prefixes to their morphological category and likely root transformation (e.g., '-ing' → present participle, '-ed' → past tense).
- 3. For each word, apply suffix-stripping rules in order of priority (longest match first).
- 4. Apply prefix-stripping rules for common English prefixes (un-, re-, pre-, dis-, mis-).
- 5. Identify the morphological features: POS tag estimation (noun, verb, adjective, adverb), tense, number, degree.
- 6. If no rule matches, the word is returned as is (base form assumption).
- 7. Evaluate results against gold standard annotations from the UD Tree Bank.

Libraries Used

- 1. nltk – for corpus access and tokenization
- 2. re – for pattern-based rule application
- 3. collections – for organizing rule dictionaries
- 4. pandas – for result tabulation and display

Test Cases

- 1. Input: 'running' → Rule: strip '-ing' → Root: 'run' → Feature: VERB, present participle
- 2. Input: 'unhappiness' → Prefix rule: 'un-' → Root after suffix '-ness': 'happy' → Feature: ADJECTIVE (negated)

- 3. Input: 'dogs' → Rule: strip '-s' → Root: 'dog' → Feature: NOUN, plural
- 4. Input: 'quickly' → Rule: strip '-ly' → Root: 'quick' → Feature: ADVERB from ADJECTIVE

Results

The rule-based morphological analyzer correctly identified roots and morphological features for regular English words. It achieved high precision on inflectional morphology (plural, tense) but struggled with irregular forms (e.g., 'ran' → 'run'). The UD Tree Bank evaluation showed approximately 78% accuracy on regular words.